

ECE4132 assignment 6: state estimation for a cart-pendulum system

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Aims

The aim of this assignment is to gain familiarity with the design and implementation of state estimators (or observers).

References and resources

- K. J. Åström and R. M. Murray *Feedback systems: an introduction for scientists and engineers* Princeton University Press
- ECE4132 week 9 lecture notes and slides.

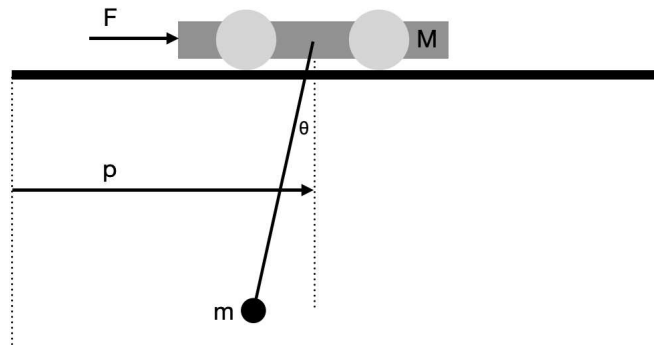
Overview

In this assignment you will

- Design a state estimator for a linearisation of a cart-pendulum system.
- Investigate how your state estimator works on the actual cart-pendulum system.
- Combine your estimator with a state feedback controller to obtain an output feedback controller for the system.

1 System model

The system we work with in this assignment consists of a crane hanging from a cart. We will assume the pendulum is in the non-inverted position. This is a reasonable model of a gantry crane, for instance. **Note: there are small differences between the model in this lab and the model in assignment 5 because of this.**



The angular displacement of the pendulum from the vertical position at time t is $\theta(t)$. The position of the cart from a reference position at time t is $p(t)$. The input to the system at time t is a force $F(t)$ acting on the cart as shown.

These variables are related by the nonlinear differential equation

$$\begin{bmatrix} (M + m) & ml \cos(\theta(t)) \\ ml \cos(\theta(t)) & (J + ml^2) \end{bmatrix} \begin{bmatrix} \frac{dp}{dt} \\ \frac{d\theta}{dt} \end{bmatrix} = \begin{bmatrix} F(t) \\ 0 \end{bmatrix} - \begin{bmatrix} c \frac{dp}{dt} - ml \sin(\theta(t)) \left(\frac{d\theta}{dt} \right)^2 \\ \gamma \frac{d\theta}{dt} + mgl \sin(\theta(t)) \end{bmatrix}.$$

The parameters of the differential equation are given below:

- Mass of pendulum: $m = 0.2$ kg
- Mass of cart: $M = 0.5$ kg
- Acceleration due to gravity $g = 9.8$ ms⁻²
- Length from pivot to center of mass: $l = 0.3$ m
- Moment of inertia about the center of mass: $J = 0.006$ kgm²
- Coefficient of friction of the cart $c = 0.1$
- Coefficient of friction of the pendulum $\gamma = 0.005$

Throughout, we make the choice of state

$$\mathbf{x}(t) = \begin{bmatrix} p(t) \\ \theta(t) \\ \frac{dp}{dt} \\ \frac{d\theta}{dt} \end{bmatrix}.$$

Using these variables, the system can be expressed in nonlinear state-space form as

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & M+m & -ml \cos(x_2(t)) \\ 0 & 0 & -ml \cos(x_2(t)) & J+ml^2 \end{bmatrix}^{-1} \begin{bmatrix} x_3(t) \\ x_4(t) \\ F(t) - cx_3(t) - ml \sin(x_2(t))(x_3(t))^2 \\ -\gamma x_4(t) + mgl \sin(x_2(t)) \end{bmatrix}$$

The system has an equilibrium point at $F = 0$ and $\mathbf{x} = \mathbf{0}$ corresponding to the cart being stationary at position zero, the pendulum being stationary and balancing in the *downward* vertical position, and there being no external force acting on the cart. If we linearise about the equilibrium point, the resulting linear system is

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}F(t) \\ y(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}F(t) \end{aligned}$$

where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & M+m & ml \\ 0 & 0 & ml & J+ml^2 \end{bmatrix}^{-1} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -c & 0 \\ 0 & -mgl & 0 & -\gamma \end{bmatrix} \quad (1)$$

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & M+m & ml \\ 0 & 0 & ml & J+ml^2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad (2)$$

$$\mathbf{C} = [1 \quad 0 \quad 0 \quad 0] \quad (3)$$

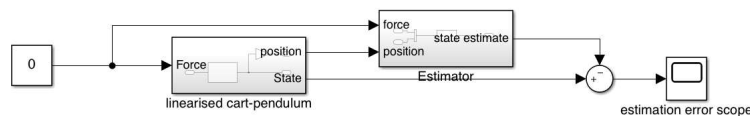
$$\mathbf{D} = 0 \quad (4)$$

Answer the following question in the Assignment 6 Moodle quiz:

Q1 Consider the linearised cart-pendulum system given above in equations (1), (2), (3), and (4).

- What is the determinant of the observability matrix of the system? (Give your answer to two significant figures.)
- Is the linearised system observable?
- If you measured the angular position of the system instead of the (linear) position, would the linearised system still be observable?

2 State estimation for the linearised model



In this section, your aim is to design the parameters of a state estimator and see how it performs when estimating the state of the linearised plant model. The state estimator has the form:

$$\frac{d\hat{\mathbf{x}}}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) + \mathbf{L}(F(t) - \mathbf{C}\hat{\mathbf{x}}(t)) = (\mathbf{A} - \mathbf{LC})\hat{\mathbf{x}}(t) + [\mathbf{B} \quad \mathbf{L}] \begin{bmatrix} F(t) \\ y(t) \end{bmatrix}$$

where $F(t)$ and $y(t)$ (the input and output of the plant) are the two inputs to the observer and $\hat{\mathbf{x}}(t)$ (the state of the observer) is also the output of the observer.

- Find a 4×1 matrix $\mathbf{L}$ so that the eigenvalues of $\mathbf{A} - \mathbf{LC}$ are at $-8 \pm 0.5j$ and $-12 \pm 0.5j$. (Hint: use `place`.)

Answer the following question in the Assignment 6 Moodle quiz:

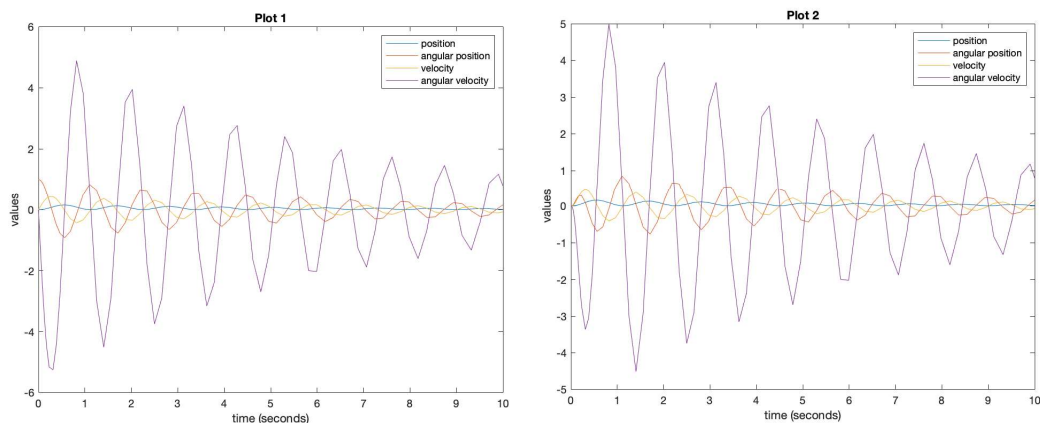
Q2 What are the entries of the matrix $\mathbf{L}$? (Specify each entry correct to 2 significant figures.)

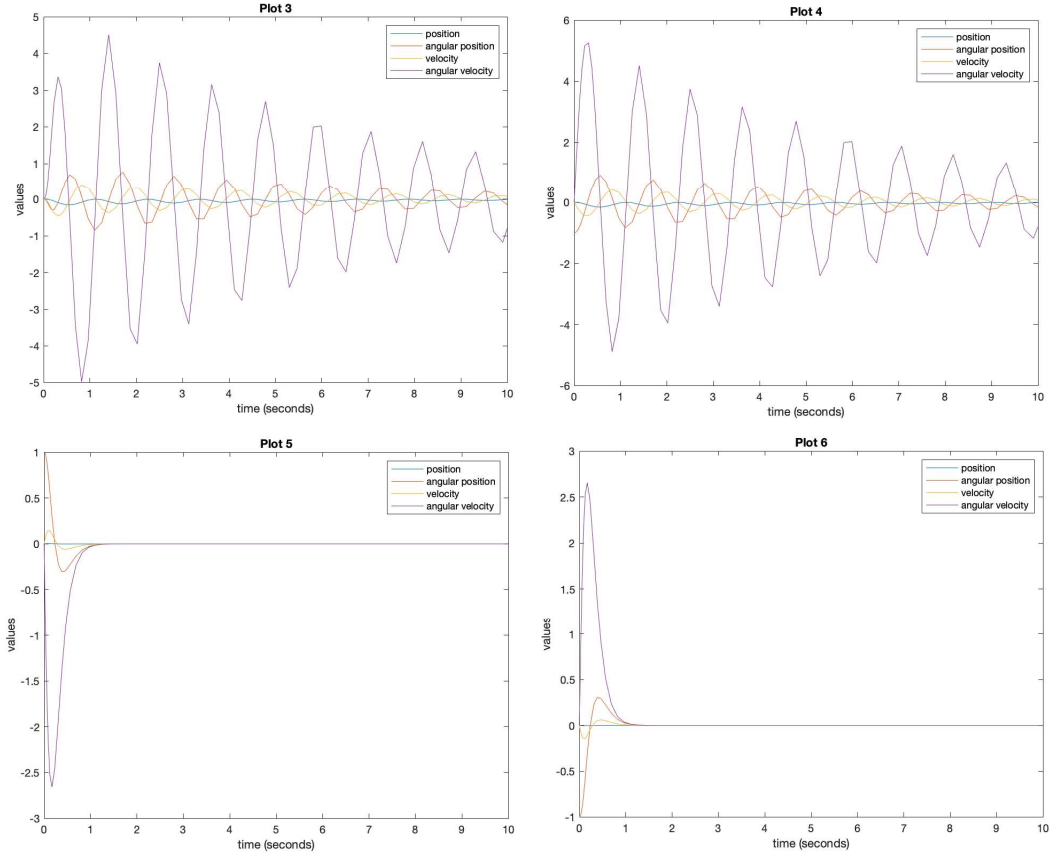
Note: the rounded values should only be used for the Moodle quiz. Use the un-rounded values of $\mathbf{L}$ in what follows.

- Set the parameters `Ahat`, `Bhat`, `Chat`, and `Dhat` in the script `setupModelParameters.m` so that the estimator state space model in the `linCartPendulumEst.slx` correctly implements your observer design. (Hint: think carefully about what $\mathbf{C}$ and $\mathbf{D}$ should be.)
- Set the initial state of the estimator, `xhatinit`, to be the zero vector.
- Set the initial state of the plant to $[0 \ 1 \ 0 \ 0]^T$. Simulate the system and plot the estimation error (the state of the plant minus the state of the estimator) for all four states on the same set of axes.
- Set the initial state of the plant to $[0 \ -1 \ 0 \ 0]^T$. Simulate the system and plot the estimation error (the state of the plant minus the state of the estimator) for all four states on the same set of axes.

Answer the following question in the Assignment 6 Moodle quiz:

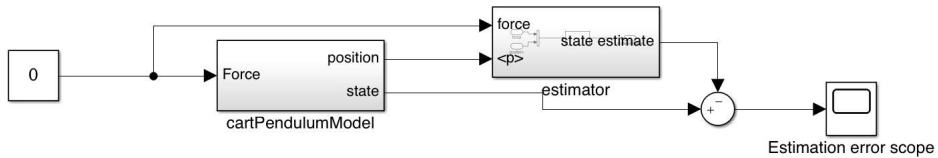
Q3 Consider the following six plots (labelled Plots 1–6, for reference).





- Which plot corresponds to the estimation error (between the linearised plant state and the estimator state) when the initial estimator state is 0 and the initial plant state is $[0 \ 1 \ 0 \ 0]^T$?
- Which plot corresponds to the estimation error (between the linearised plant state and the estimator state) when the initial estimator state is 0 and the initial plant state is $[0 \ -1 \ 0 \ 0]^T$?

3 State estimation for the nonlinear system

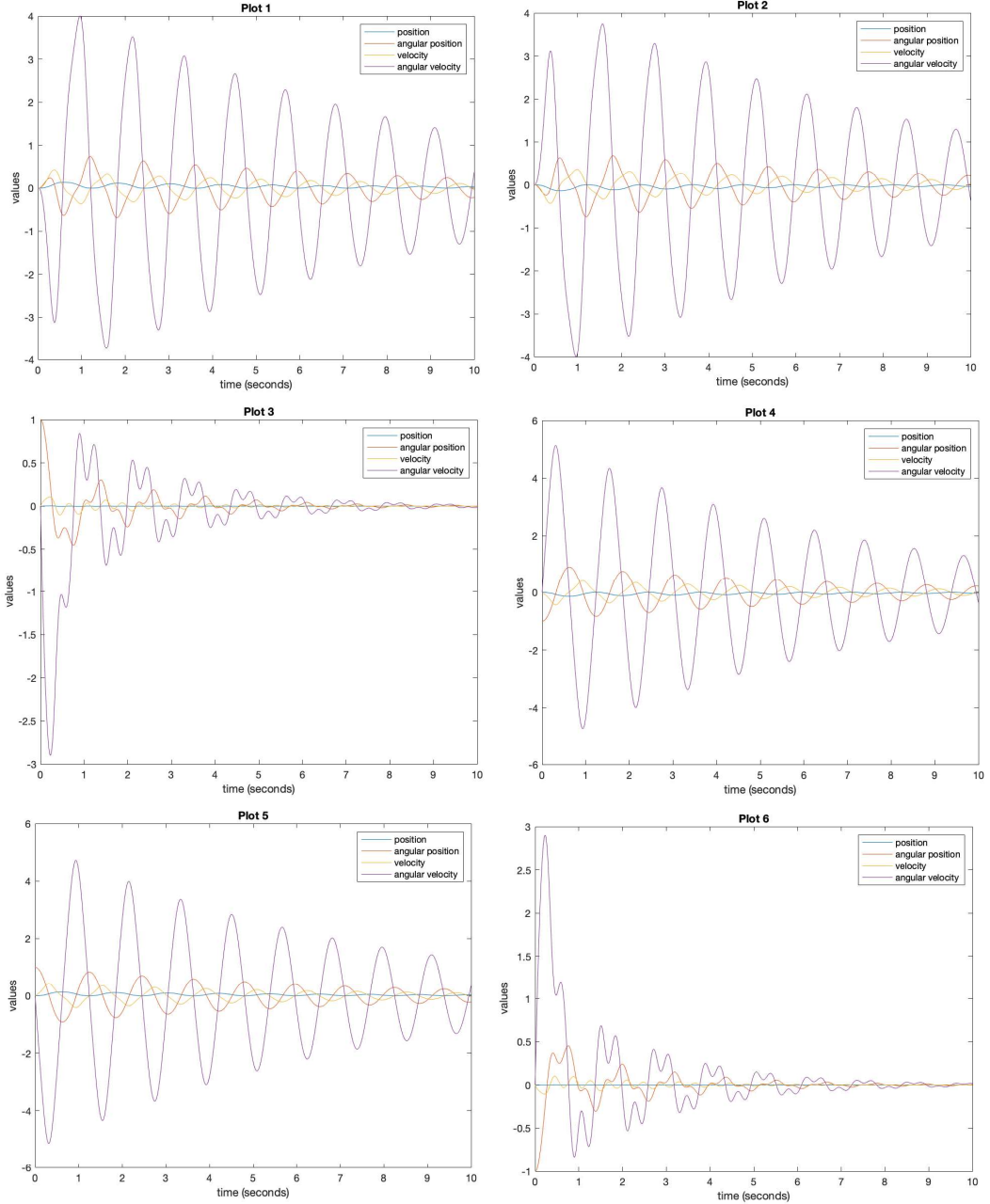


- Open the Simulink model `cartPendulumEst.slx`.
- Add a state estimator of the same form that you used in Section 2 to your simulink model. The two inputs to the estimator should be the input $F(t)$ to the plant and the output $y(t)$ of the plant. The output of the estimator should be the estimated state $\hat{\mathbf{x}}(t)$
- In `setupModelParameters.m`, ensure that the parameters of the estimator A_{hat} , B_{hat} , C_{hat} , and D_{hat} and the initial state `xhatinit` are set the same as in Section 2.
- Set the initial state of the plant to $[0 \ 1 \ 0 \ 0]^T$. Simulate the system and plot the estimation error (the state of the plant minus the state of the estimator) for all four states on the same set of axes.

- Set the initial state of the plant to $[0 \ -1 \ 0 \ 0]^T$. Simulate the system and plot the estimation error (the state of the plant minus the state of the estimator) for all four states on the same set of axes.

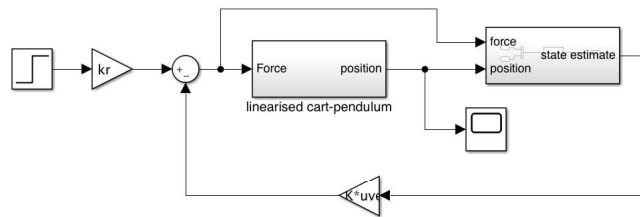
Answer the following question in the Assignment 6 Moodle quiz:

Q4 Consider the following six plots (labelled Plots 1–6, for reference).



- Which plot corresponds to the estimation error (between the nonlinear plant state and the estimator state) when the initial estimator state is 0 and the initial plant state is $[0 \ 1 \ 0 \ 0]^T$?
- Which plot corresponds to the estimation error (between the nonlinear plant state and the estimator state) when the initial estimator state is 0 and the initial plant state is $[0 \ -1 \ 0 \ 0]^T$?

4 Output feedback for the linear system



- Open the Simulink model `linCartPendulumOF.slx`.
- Connect up the state estimator (which is of the same form that you used in Section 2) to your simulink model. In `setupModelParameters.m`, ensure that the parameters of the estimator `Ahat`, `Bhat`, `Chat`, and `Dhat` and the initial state `xhatinit` are set the same as in Section 2.
- Add a controller of the form $F(t) = k_r r(t) - \mathbf{K}\hat{\mathbf{x}}(t)$ to your Simulink model.
- Design a 1×4 matrix of state feedback gains $\mathbf{K}$ so that the closed-loop eigenvalues of $\mathbf{A} - \mathbf{BK}$ are at $-4 + \pm 0.5j$ and -5 and -6 .

Answer the following question in the Assignment 6 Moodle quiz:

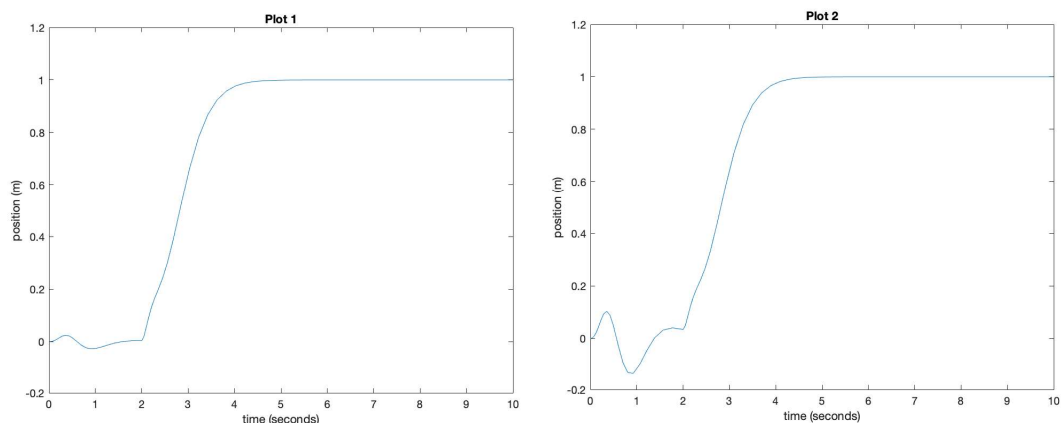
Q5 What are the entries of the matrix $\mathbf{K}$? (Specify each entry correct to 2 significant figures.)

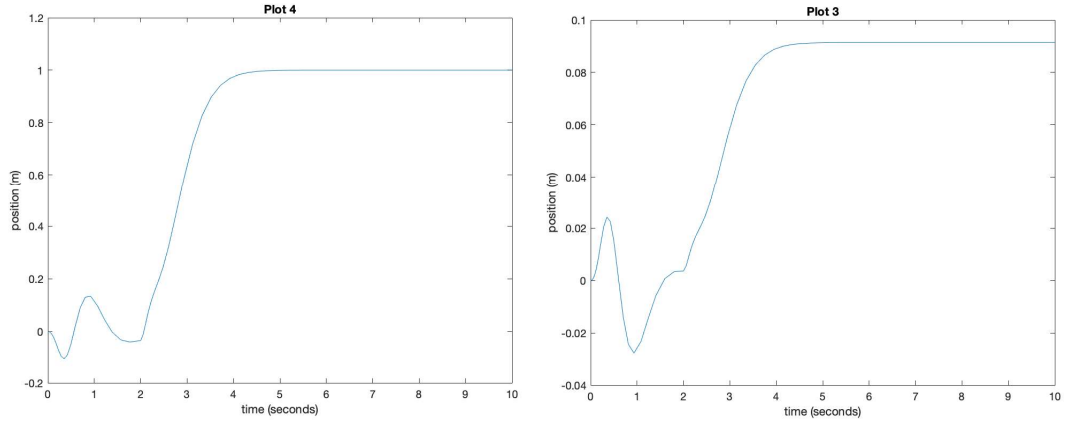
Note: the rounded values should only be used for the Moodle quiz. Use the un-rounded values of $\mathbf{K}$ in what follows.

- Choose a value of the reference gain k_r so that the steady-state value of the position output is the same as the steady-state value of the reference input.
- Set the reference input to step from 0 to 1 at time $t = 2$.
- Set the initial state of the plant to $[0 \ 0.2 \ 0 \ 0.2]^T$. Simulate the system and plot the output (position) vs time when the input is a step reference that steps from 0 to 1 at time $t = 2$ seconds.

Answer the following question in the Assignment 6 Moodle quiz:

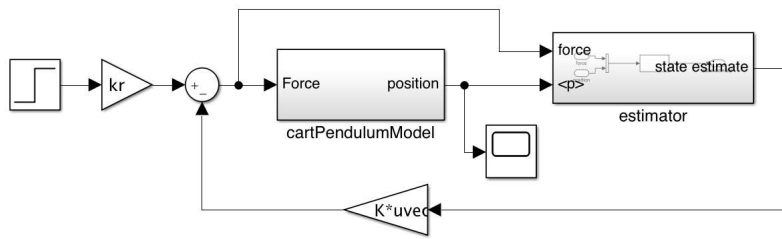
Q6 Consider the following four plots (labelled Plots 1–4, for reference).





Which plot corresponds to a plot of the output of the linearised plant (position) vs time when the input is a step reference that steps from 0 to 1 at time $t = 2$ seconds and the initial state is $[0 \ 0.2 \ 0 \ 0.2]^T$?

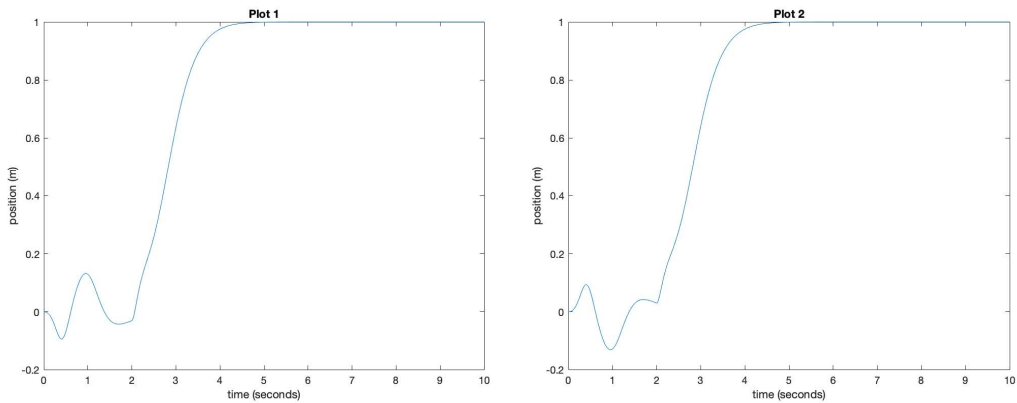
5 Output feedback for the nonlinear system

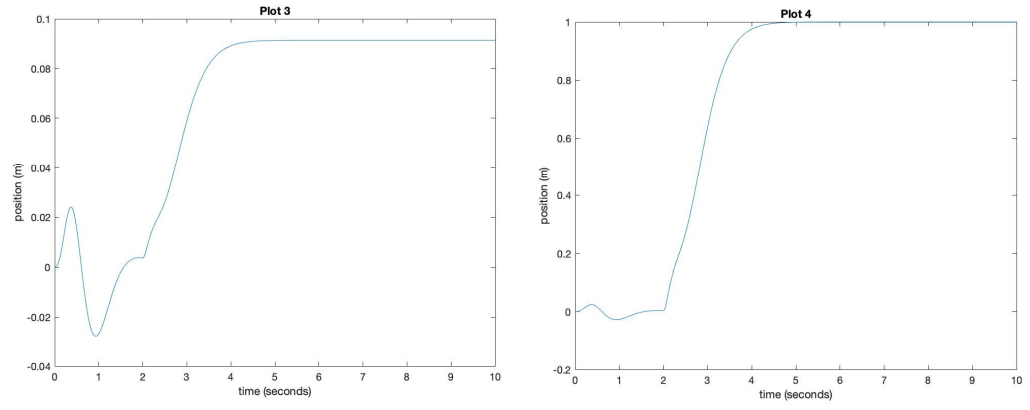


1. Open the Simulink model `CartPendulumOF.slx`.
2. Add a controller of the form $F(t) = k_r r(t) - \mathbf{K} \hat{\mathbf{x}}(t)$ to your Simulink model.
3. Use the same matrix $\mathbf{K}$ of feedback gains and reference gain k_r as in Section 4.
4. Set the initial state of the plant to $[0 \ 0.2 \ 0 \ 0.2]^T$. Simulate the system and plot the output (position) vs time when the input is a step reference that steps from 0 to 1 at time $t = 2$ seconds.

Answer the following question in the Assignment 6 Moodle quiz:

Q7 Consider the following four plots (labelled Plots 1–4, for reference).





Which plot corresponds to a plot of the output of the nonlinear system (position of the crane) vs time when the input is a step reference that steps from 0 to 1 at time $t = 2$, and the initial state is $[0 \ 0.2 \ 0 \ 0.2]^T$?